

IKT215 Exercise 9:

Error Estimation Rules for Classification

March 26th, 2026

Exercise sessions, while not mandatory, are an integral part of your learning experience and exam preparation. Some exam questions will be based on tasks covered during these sessions. Although you have the flexibility to complete exercises at home and ask questions later (preferably within one week), I strongly recommend attending the sessions in person for immediate support and guidance. This allows you to benefit from direct interaction and real-time problem-solving. I respond promptly to emails (manuel.s.mathew@uia.no).

Task 1: Test-Set Error Estimation

Implement a holdout test-set error estimation.

Dataset: Breast Cancer Dataset (569 samples, 30 features, binary classification).

1. Load the dataset using `sklearn.datasets.load_breast_cancer()`.
2. Manually shuffle and split the dataset into 80% training and 20% testing using indices.
3. Train a decision tree classifier (`max_depth=2`, `splitter="random"`) and compute training/test accuracies.

Questions:

1. Report training and test accuracies.
2. Why does the test-set error provide a more realistic performance estimate than the train error?

Task 2: Resubstitution Error Estimation

Compare resubstitution error with test error using a 1-NN classifier.

1. Train a 1-NN classifier on the training set from Task 1.
2. Compute resubstitution error (training accuracy) using the training set.
3. Evaluate the model on the test set from Task 1.

Questions:

1. Report resubstitution and test accuracies.
2. Why does 1-NN achieve 100% training accuracy but poor test performance?
3. Which of the two errors is a more accurate estimation (closer to true error ϵ_n) of the classifier performance?

Task 3: Cross-Validation Error Estimation

Implement 5-fold CV from scratch.

1. Split data into 5 equal folds using indices.
2. Use a decision tree classifier (`max_depth=2`, `splitter="random"`)
3. Train/test the model on each fold combination.
4. Compute mean accuracy and standard deviation.

Questions:

1. Report mean CV accuracy and standard deviation.
2. Why does cross-validation accuracy differ from test-set accuracy?
3. Why does cross-validation reduce variability compared to a single train-test split?